

HAT-P-54b

R-1676

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_181024 · created 2026-08-02T11:04:14+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458415.82041 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.025
Transit depth (Rp/Rs)^2	2.86 +/- 0.86 [%]
Orbital Inclination (inc)	89.88 +/- 1.46
Airmass coefficient 1 (a1)	1.051 +/- 0.032
Airmass coefficient 2 (a2)	-0.0303 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	3.03 %
Best Comparison Star	#2 - [536, 385]
Optimal Method	PSF photometry
Transit Duration (day)	0.0732 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.169 ± 0.025 vs 0.1572 (deviation 0.47σ)
Duration fit vs published	0.0732 d vs 0.075 d (deviation 0.78σ)
Mid-time O-C	-3.8 min
Depth SNR	6.8
Residual scatter	3.03 %

Light curve

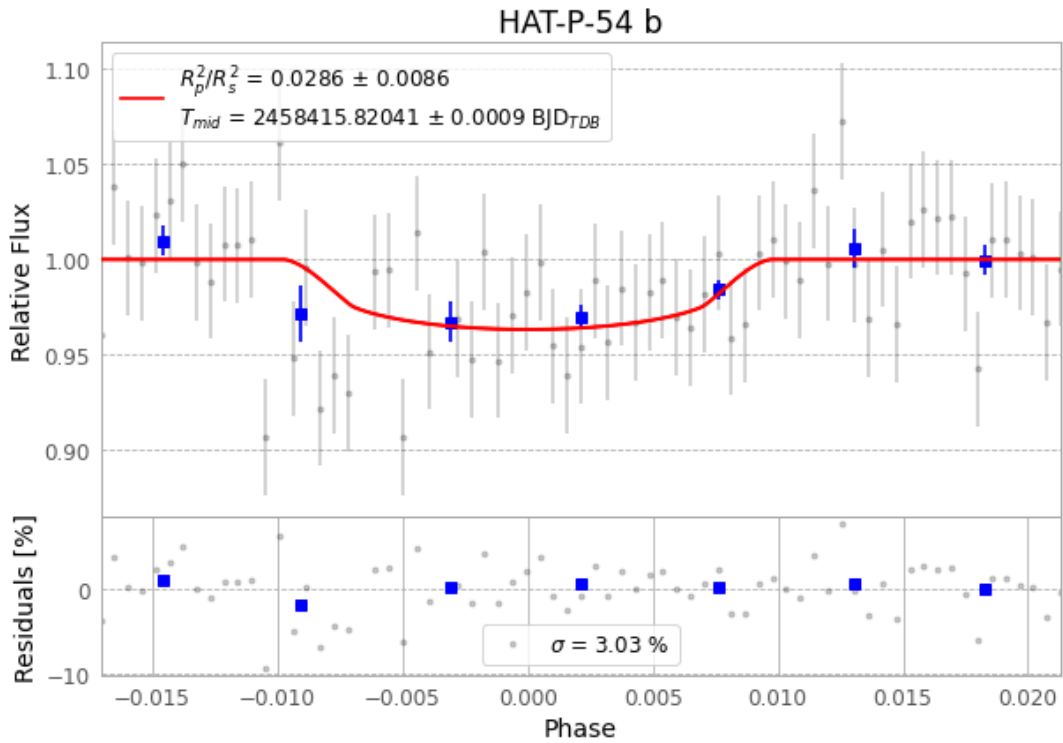


Figure 1 — FinalLightCurve_HAT-P-54 b_2018-10-23.png (SHA-256 0c15379d4bac328a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 138], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20b6fa747c694156...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), HATP-54181024060313.FITS → HATP-54181024093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-181024.log (313d54c811ea...), FinalLightCurve_HAT-P-54 b_2018-10-23.png (0c15379d4bac...), FinalLightCurve_HAT-P-54 b_2018-10-23.csv (9afcca90685d...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 11:04Z.